Per−copy profile HMMs + EM priors on full−length FLNC−like reads

Toy example sized for FLNC (full−length reads, multi−SNP discrimination). HMM alone hard−calls the clear−margin reads; EM uses those calls to soft−assign the rest.

Per−copy HMMs + EM priors − toy example representative of FLNC

3 copies of length 15, 4 SNP positions. Reads are FULL−LENGTH (cover the entire transcript), 8% per−base error.

```
Copy A: TCCGCTGTTGGCGCT
Copy B: TCGGCTCTTGTCGTT
Copy C: TCTGCTTTTGACGGT
```

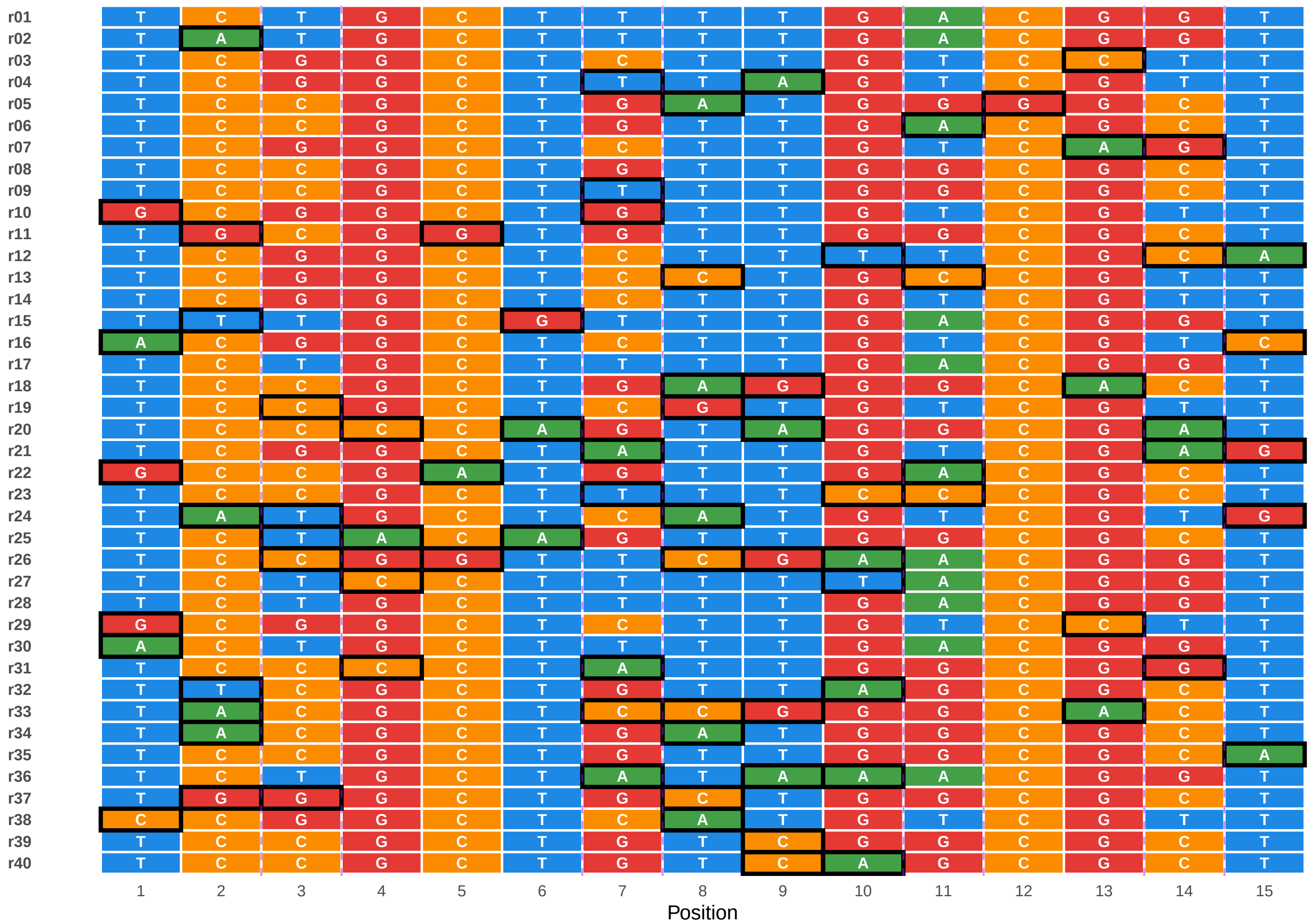
SNPs at positions 3, 7, 11, 14. Other positions are shared across copies.

True abundances: theta_A = 0.50, theta_B = 0.30, theta_C = 0.20

Library: 40 FLNC reads simulated.

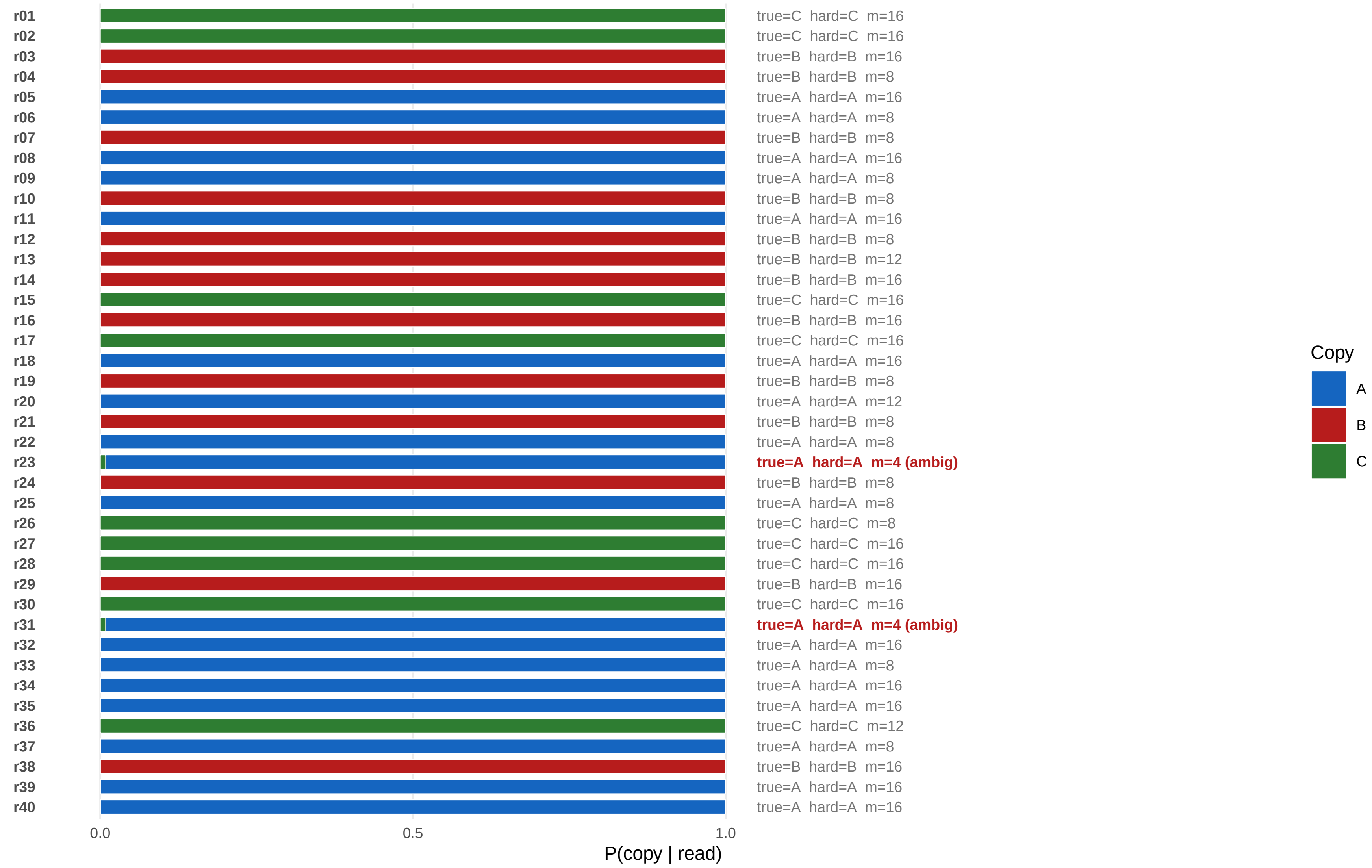
40 full−length FLNC reads − base−by−base view

Purple dashed lines bracket the 4 SNP positions. Black border = position where the read differs from its true source (= error).



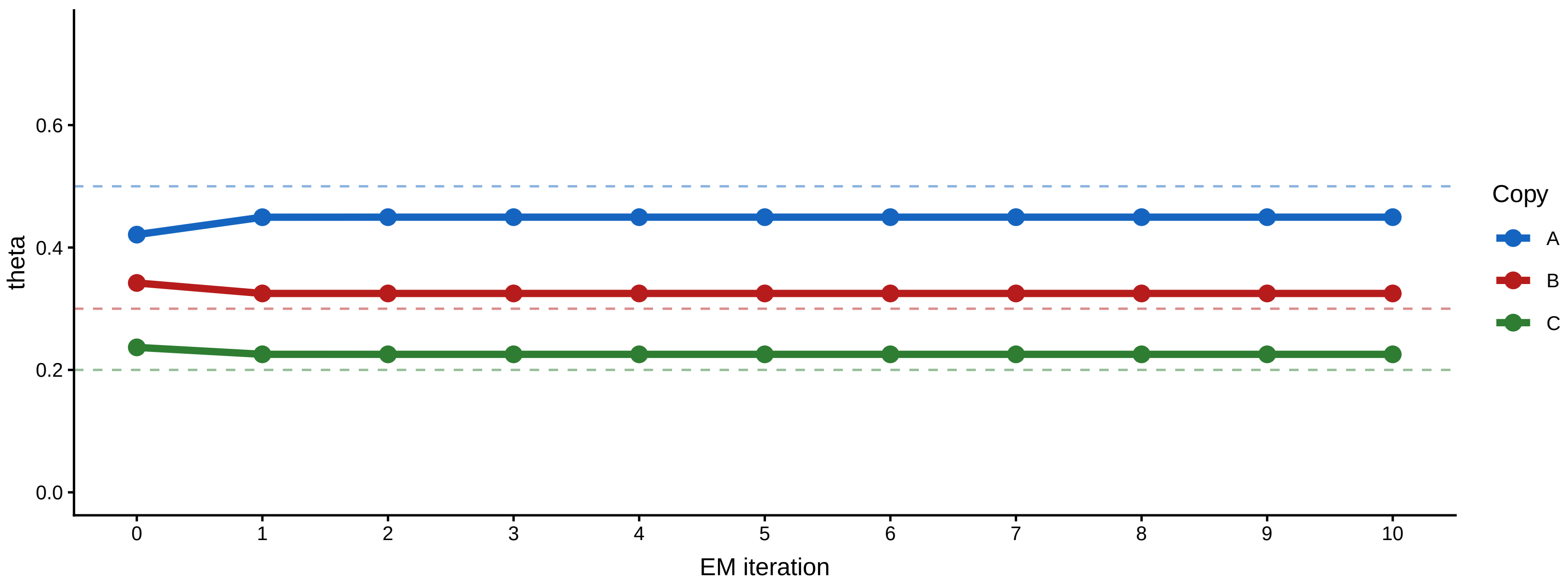
Per−read posterior after EM

Stacked bar = P(A|r), P(B|r), P(C|r). Right text = true source, HMM hard call, DlogL margin.



EM trajectory of theta

Iter 0 = bootstrap (38 / 40 clean reads). Dashed = true theta. Converges quickly.



True vs estimated theta